

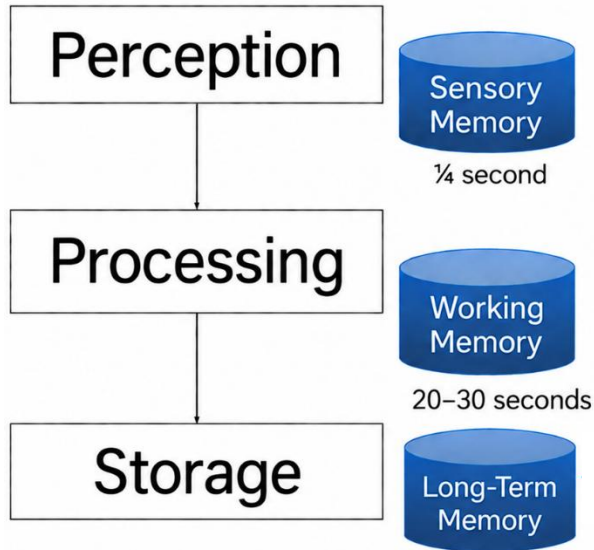
Chapter 3: Learning through exploration

Digital Education (CS-411)

September 22, 2026

From memory to learning

Information

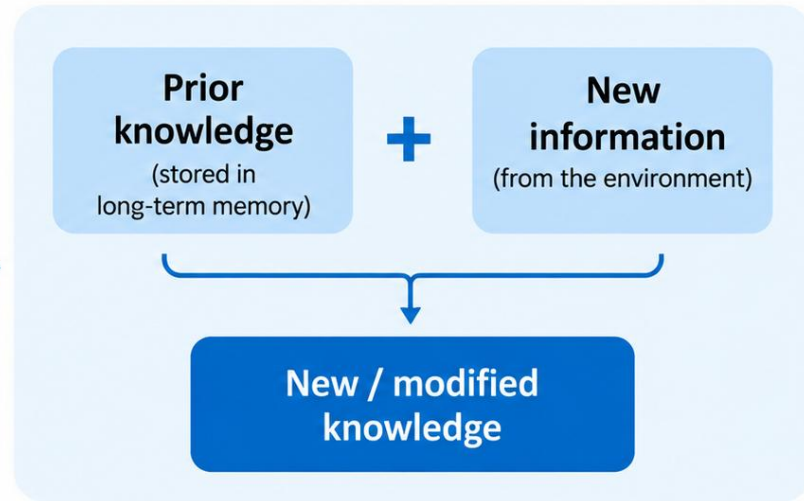
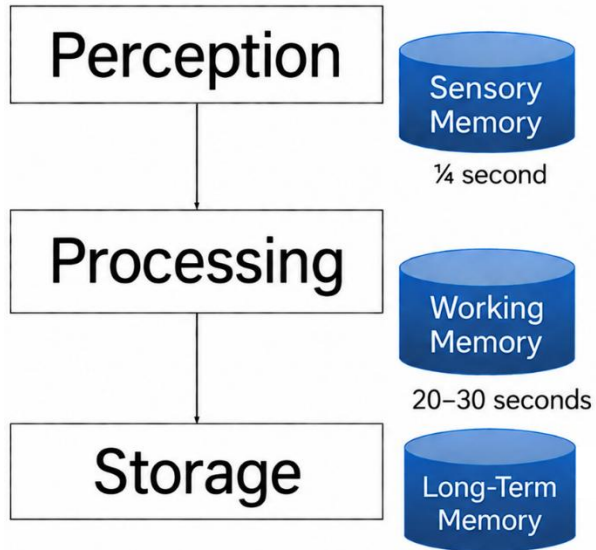


Knowledge

- Learning comes from processing information perceived or stored in long-term memory
- Processing information is managed by the working memory

Working memory taps into long-term memory

Information



Knowledge

Learning fractions

Go maths.ch Le site d'entraînement au calcul mental et autres techniques de calcul

Lundi 21 septembre Accueil

Calcul mental
Livrets
Compléments
Divers

Les 4 opérations
Addition
Soustraction
Multiplication
Division

Pour les plus grands
Priorités des opérations
Entiers relatifs
Fractions
Changements d'unités
Pourcents
Algèbre
Et tout le reste...

Code fractionnaire - les 4 opérations

Tu exerces ici un mélange des 4 opérations en code fractionnaire. Comme toujours, les réponses sont attendues en code irréductible !
Si tu le souhaites, tu peux amplifier ou simplifier directement à l'écran les fractions initiales.

Dénominateur max: 10 | les quatre opérations | entraînement

Commencer Voir la réponse <

$__$ $__$ = $\frac{__}{__}$ Vérifie

✓ ✗

Imprimer une fiche d'exercices

https://gomaths.edu-vd.ch/cf_mix.php

mixed fraction

										show	hide
1										1	
$\frac{1}{2}$										$\frac{1}{2}$	
										0	
					$\frac{1}{8}$				$\frac{1}{8}$	$\frac{2}{8}$	
$\frac{1}{3}$										$\frac{1}{3}$	
										0	
										0	
										0	
										0	

<https://www.visnos.com/demos/fraction-wall>

Gomaths.ch Le site d'entraînement au calcul mental et autres techniques de calcul

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Calcul mental
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Les 4 opérations
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Dénominateur max: 10 | les quatre opérations | entraînement

Commencer Voir la réponse <

— — = $\frac{\square}{\square}$

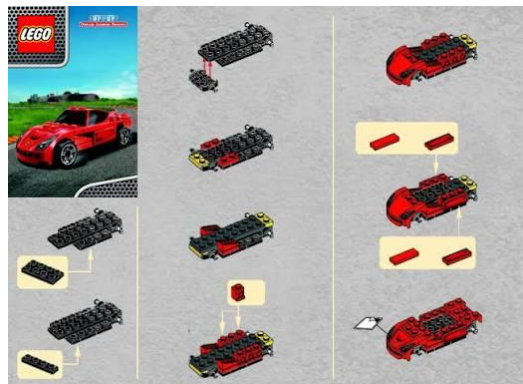
Vérifie

✓ ✗

Imprimer une fiche d'exercices



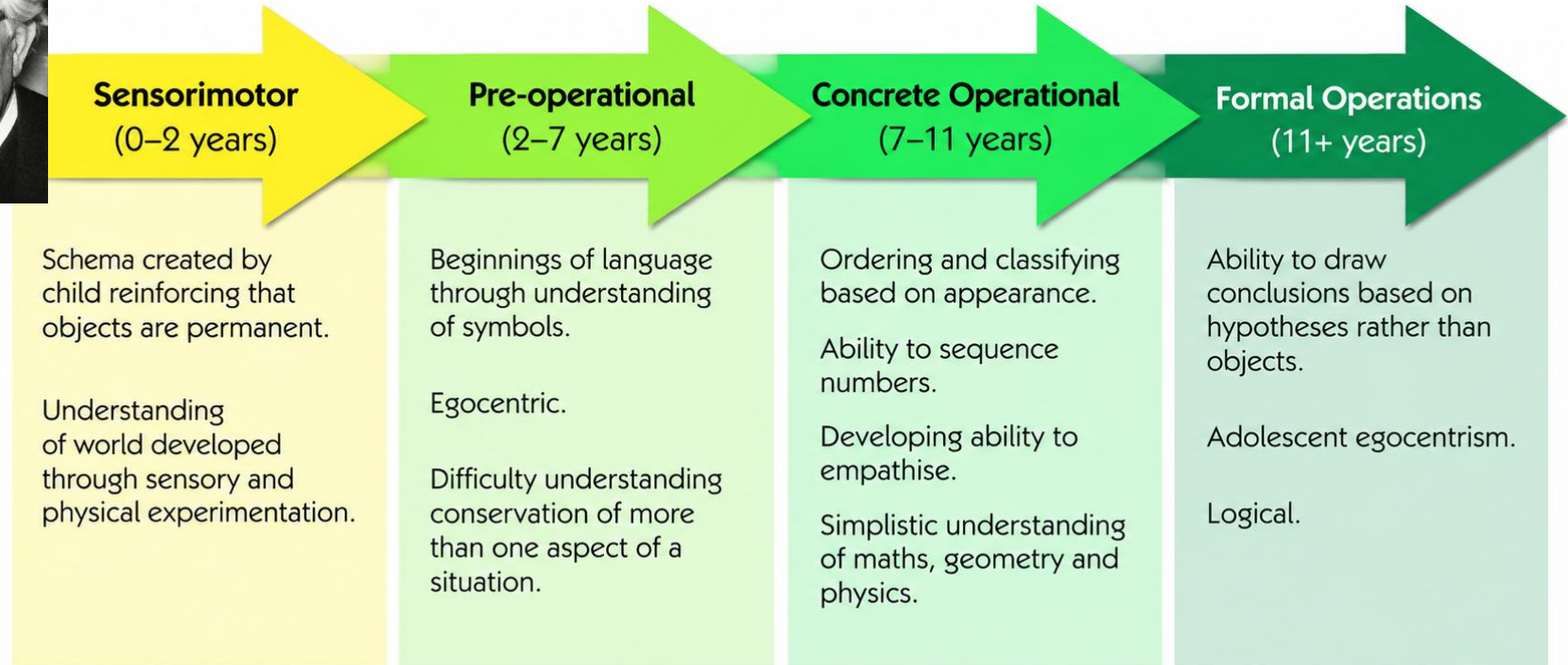
	mixed	fraction					Eq		LCM	Advanced		show	hide
1												1	
$\frac{1}{2}$												$\frac{1}{2}$	
												0	
						$\frac{1}{8}$		$\frac{1}{8}$					$\frac{2}{8}$
$\frac{1}{3}$												$\frac{1}{3}$	
												0	
												0	
												0	
												0	
												0	



Stages of development



Jean Piaget



<https://www.youtube.com/watch?v=TRF27F2bn-A>

Constructivism



Jean Piaget

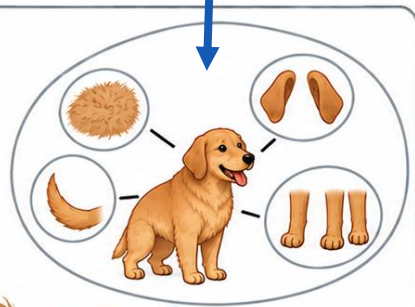
Individuals actively build knowledge from their experiences and interactions with the environment rather than passively absorbing information

Schema: The basic building blocks of knowledge or mental models used to understand the world.

Assimilation: Fitting new information into an existing schema.

Accommodation: Changing an existing schema or creating a new one when new information does not fit old ideas.

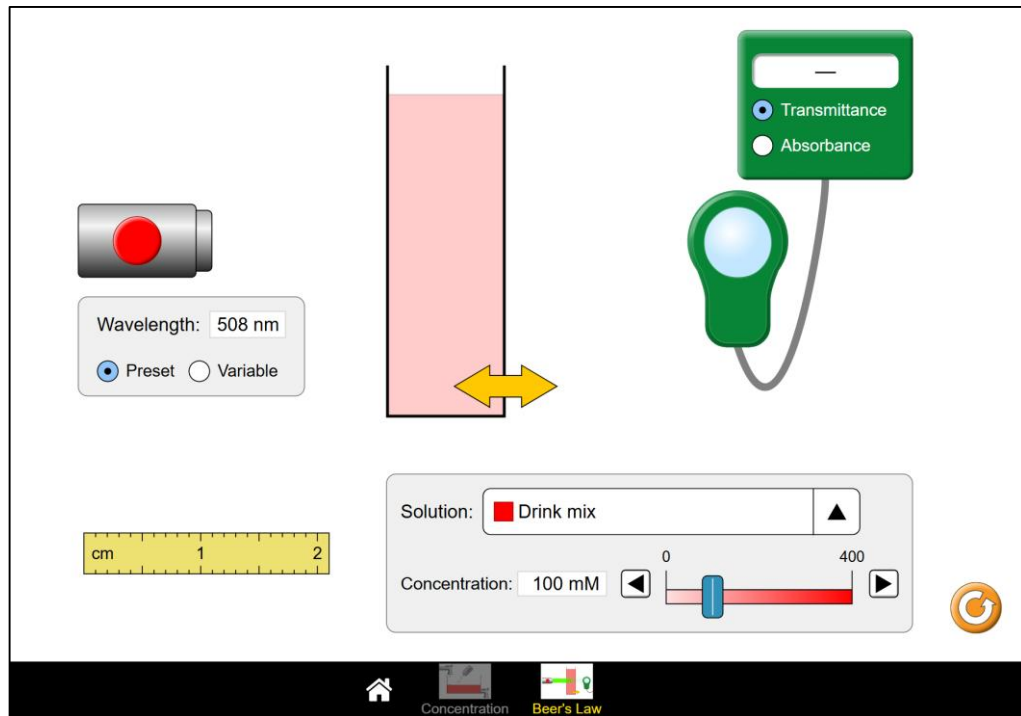
Schema



Cognitive conflict as a key learning mechanism

- Learning from experience
- Learning by doing
- Discovery learning

Explore the simulation. Try to figure out how the system works.



https://phet.colorado.edu/sims/html/beers-law-lab/latest/beers-law-lab_all.html

How can we design activities in
which students actively
construct knowledge?

From exploration to consolidation

Exploration

- look at experimental results
- compare different conditions
- notice patterns / differences
- try to explain what happened
- make predictions / hypotheses
- discuss interpretations

Consolidation

- introduce theoretical concepts
- relate the concepts to the observed patterns



There is a time for telling

Just-in-time Telling

Core learning mechanism:

Students **explore a problem or phenomenon before** receiving the relevant explanation.

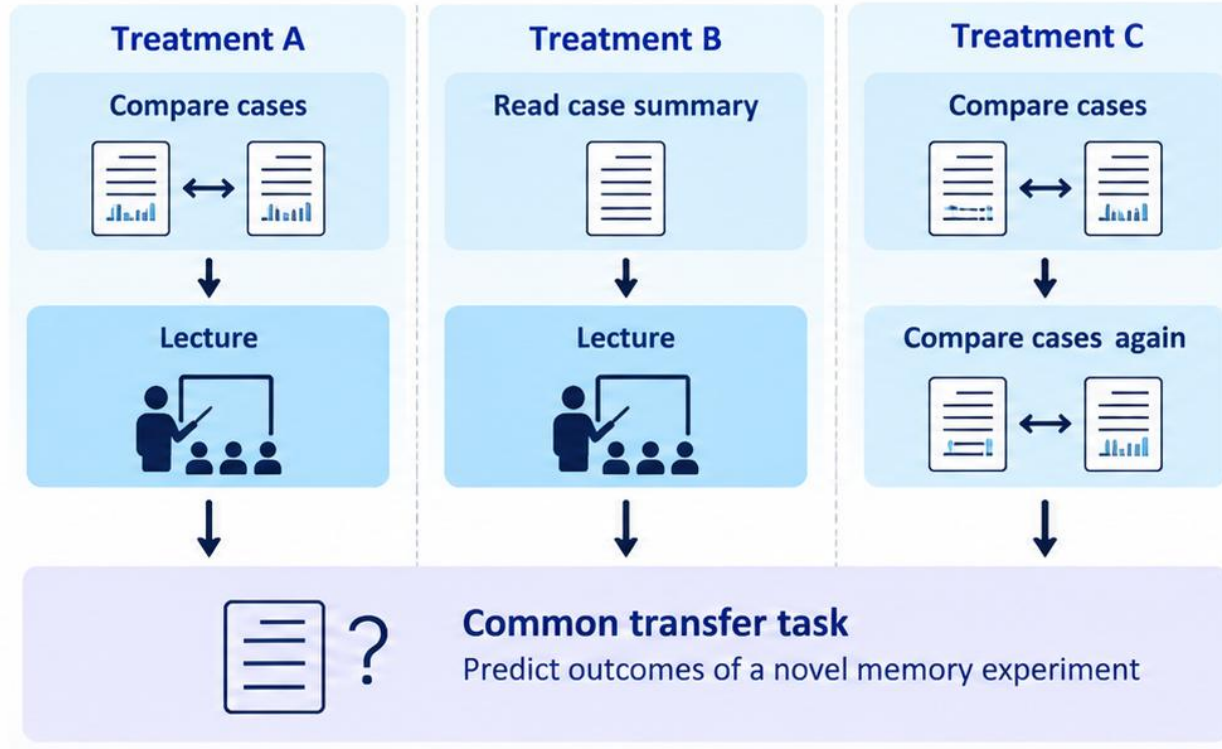
Why it works:

People learn from expository materials by integrating the explanations with their prior knowledge. But learners may not have sufficient prior knowledge to do this meaningfully. Exploring the problem first helps develop the knowledge that makes the subsequent explanation meaningful.

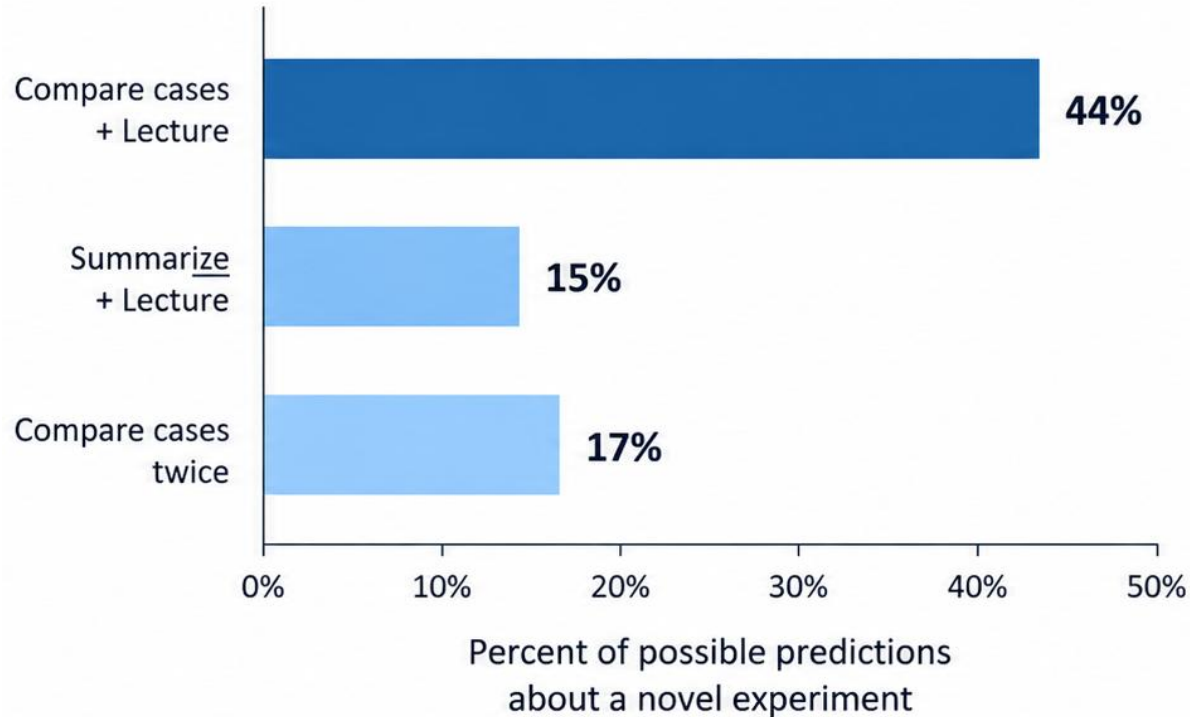
Example:

Exploring the *doctor visit* and *balloon passage* experiments before learning the underlying memory concepts.

Example: Just-in-time Telling

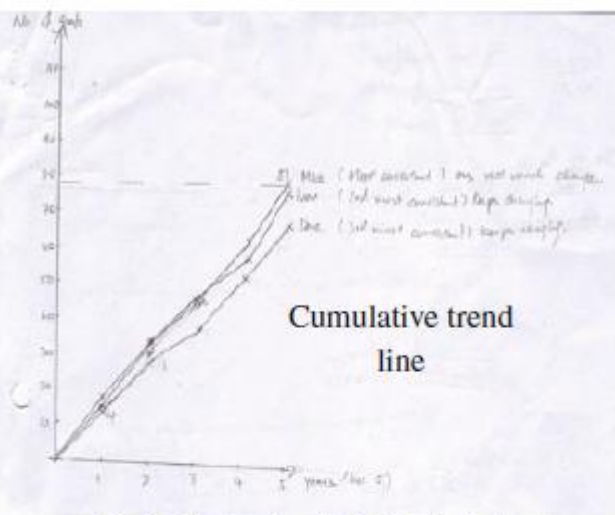
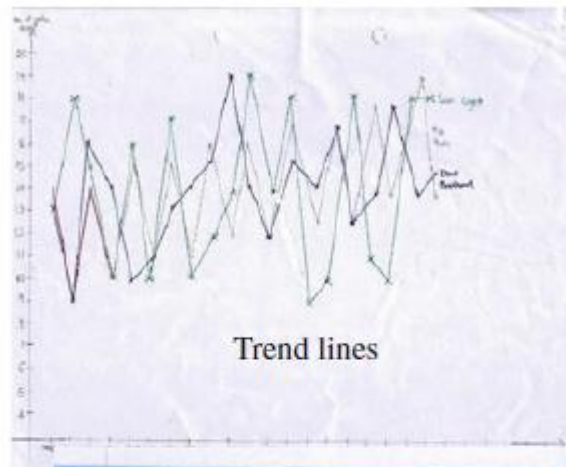


Example: Just-in-time Telling



Who is the most consistent striker?

Year	Mike Arwen	Dave Backhand	Ivan Right
1988	14	13	13
1989	9	9	18
1990	14	16	15
1991	10	14	10
1992	15	10	16
1993	11	11	10
1994	15	13	17
1995	11	14	10
1996	16	15	12
1997	12	19	14
1998	16	14	19
1999	12	12	14
2000	17	15	18
2001	13	14	9
2002	17	17	10



Mike Arwen : $\text{Mean} = \frac{280}{20}$
 $= 14 \text{ goals / year}$
 Mode = 14

Dave Bockhand : $\text{Mean} = \frac{280}{20}$
 $= 14 \text{ goals / year}$
 Mode = 14

Ivan Right : $\text{Mean} = \frac{280}{20}$
 $= 14 \text{ goals / year}$
 Mode = 18 and 10

9	10	11	12	13	14	15	16	17	18	19
1	1	2	2	2	4	2	2	2	1	1
1	1	1	1	3	6	3	1	1	1	1
1	5	1	1	1	2	1	1	1	5	1

from Question paper: Average = $\frac{280}{20}$

Mike has 8 years < average

4 years = average

8 years > average

Frequency of years above,
below, and at average

Dave has 7 years < average

6 years = average

7 years > average

Ivan has 9 years < average

2 years = average

9 years > average

Range
Mike
Dave
Ivan
Range (D1)

Mike Arwen: 9 - 17 = 8
Dave Randal: 9 - 19 = 10
Ivan Rial: 9 - 19 = 10

Sum of deviations about the mean (D4)

Year	Ave	MA	S-B	S-R	x
1983	14	14	13	13	0 -1 -1
1989	14	9	14	18	-5 -5 4
1990	14	14	16	15	0 +2 +1
1991	14	10	14	10	-4 0 -4
1992	14	15	10	16	+1 -4 +2
1993	14	11	11	10	-3 -3 -4
1994	14	15	13	17	+1 -1 +3
1995	14	11	14	10	-3 0 -4
1996	14	16	15	12	+2 +1 -2
1997	14	12	14	14	-2 +5 0
1998	14	16	14	16	+2 0 +5
1999	14	12	12	14	-2 -2 0
2000	16	17	15	18	+2 +1 +4
2001	14	13	14	9	-1 0 -5
2002	14	12	17	10	+2 +3 -4
2003	14	13	13	18	-1 -1 +4
2004	14	18	14	11	-7 0 -3
2005	14	14	18	10	0 +4 -4
2006	14	19	14	18	+5 0 +4
2007	14	14	15	13	0 +1 -4

Sum of year-on-year deviation (D2)

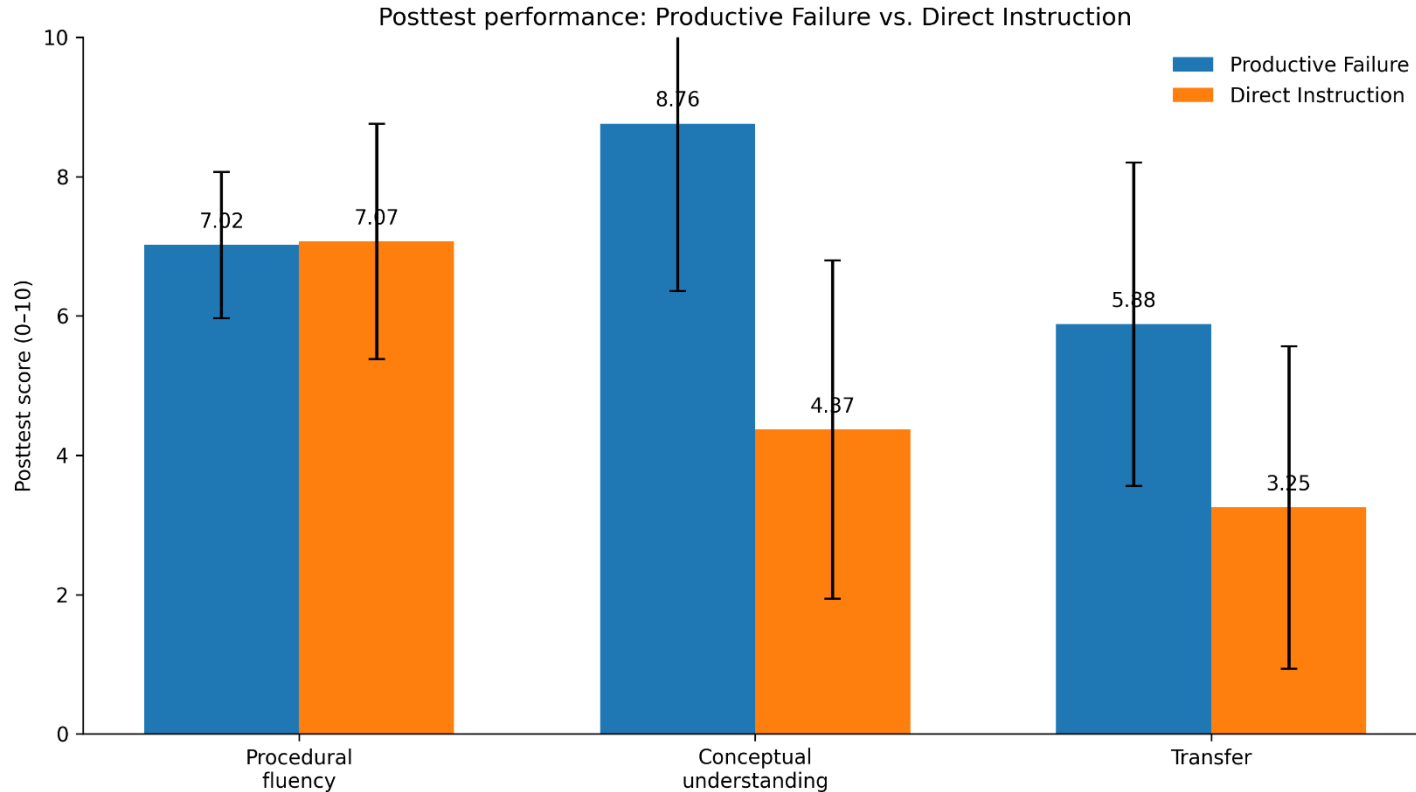
Mike: 9-14 = -5
14-9 = 5
10-14 = -4
15-10 = 5
-4
4
-4
5
-4
4
-4
5
-4
4
-4
5
-4
5
-4
0 ✓ Mike

Dave: -4
7
-2
-4
1
2
1
1
4
-5
-2
3
-1
3
-4
1
4
-4
1
-2

Ivan: 5
-3
-5
1
-6
7
-7
2
2
5
-5
4
-9
1
8
-7
-1
6
0
-5

None of the groups discovered the
canonical formula.

Productive Failure



From exploration to consolidation

Exploration

Consolidation



There is a time for telling

Productive Failure

Attempting (and failing) to generate solutions can prepare students to learn from subsequent instruction.

From exploration to consolidation

Exploration

Consolidation



How to guide learners?



There is a time for telling

Guiding learners during exploration

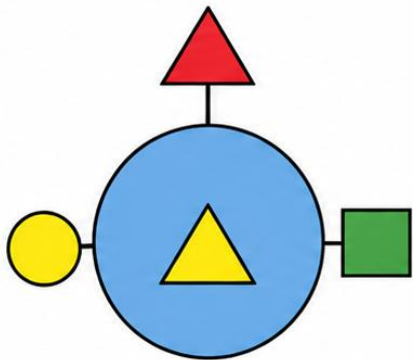
Exploration

- Carefully design the materials →

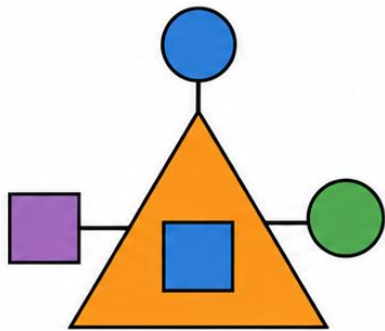
Consolidation

Year	Mike Arwen	Dave Backhand	Ivan Right
1988	14	13	13
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1991	10	14	10
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1999	12	12	14
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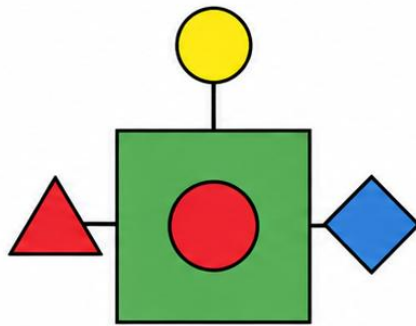
What is a Splountz?



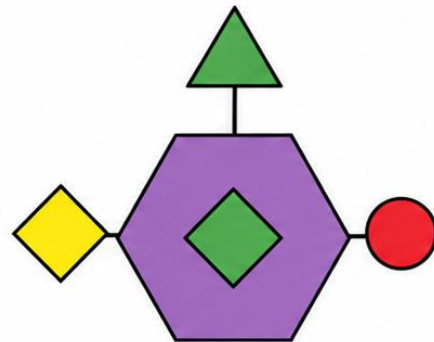
Example 1



Example 2

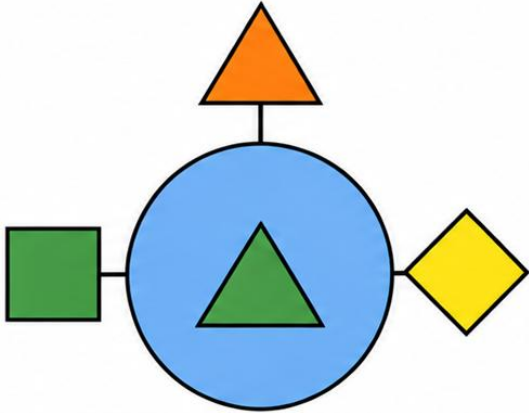


Example 3

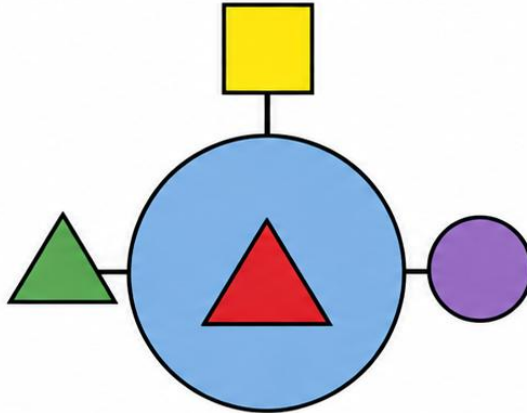


Example 4

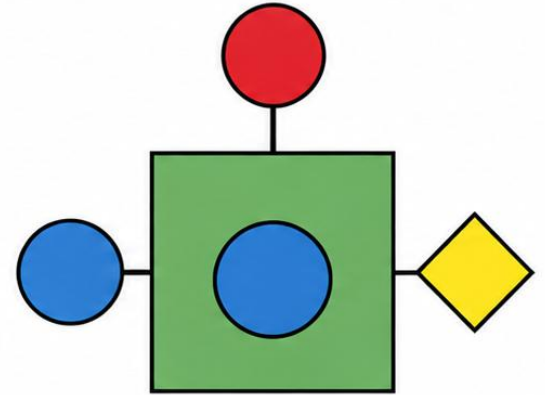
Is this a Splountz?



Yes

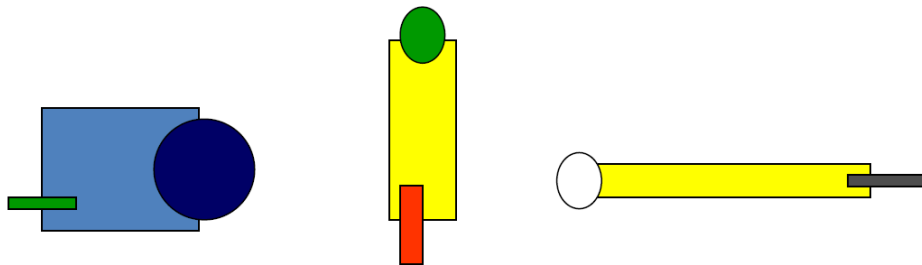


No

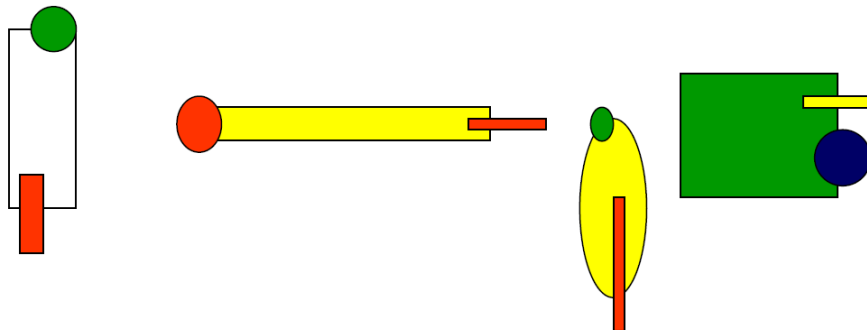


No

What is a SPUC?



Positive
instances



Negative
instances

Contrasting Cases

Core learning mechanism:

Comparing examples that differ in important ways helps learners notice which features and relationships matter.

Why it works:

People learn patterns in sensation by discerning what makes one thing different from another. Contrasting cases juxtapose “near misses” to help students pick out distinctive features.

Example:

The positive, negative, and near miss examples from the SPUC.

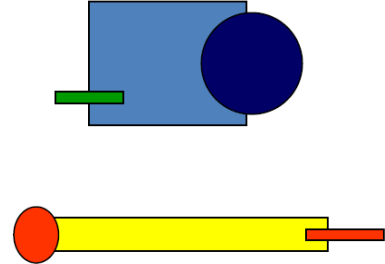
The different experiment examples in the *doctor visit* and *balloon passage*.

Guiding learners during exploration

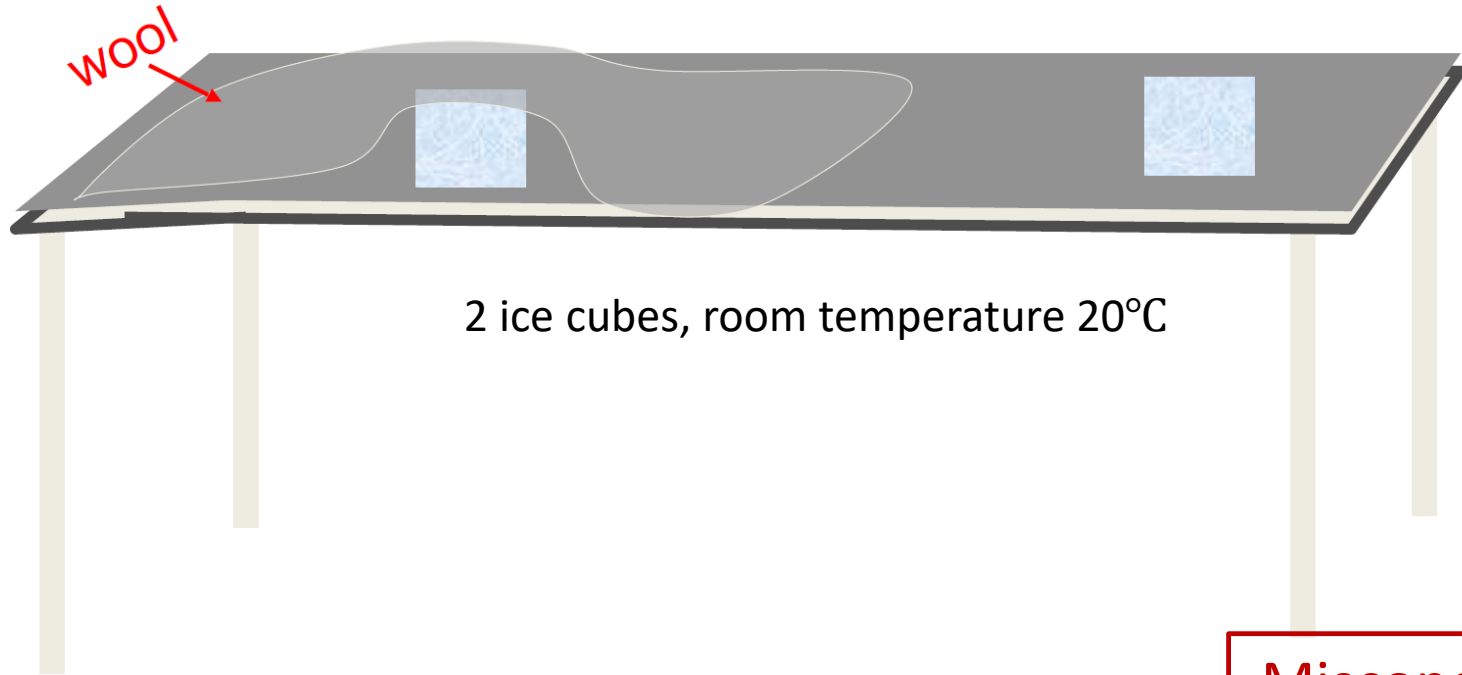
Exploration

Consolidation

- Carefully design the materials
- Comparisons, playing with differences →



Which ice cube melts faster?



Misconception

Which is more probable?

Linda is thirty-one years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and she also participated in antinuclear demonstrations. Which is more probable:

- a) Linda is a bank teller.
- b) Linda is a bank teller and is active in the feminist movement.

Heuristic

Undoing

Core learning mechanism:

Undoing is the process of identifying and replacing misconceptions and faulty methods of reasoning.

Why it works:

Misconceptions and faulty reasoning can become entrenched and difficult to detect. Undoing identifies the underlying problem and strengthens a competing mental representation that can replace the erroneous one.

Example:

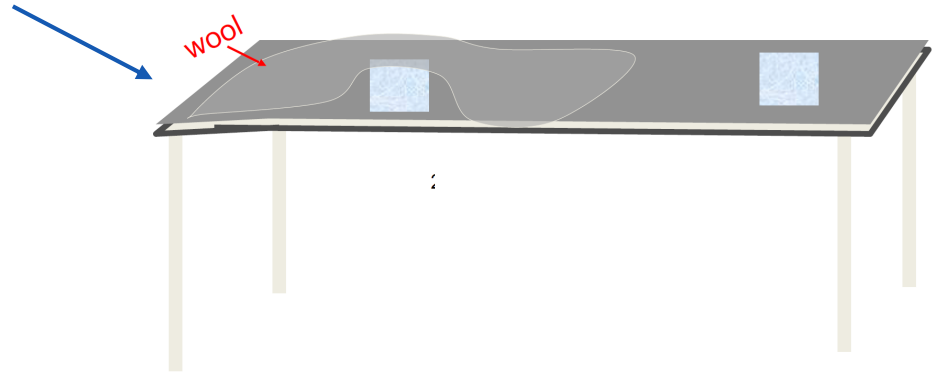
Ask learners to make a prediction on a task designed to reveal a common misconception.

Guiding learners during exploration

Exploration

Consolidation

- Carefully design the materials
- Comparisons, playing with differences
- Conflict: trapping predictions



Guiding learners during exploration

Exploration

Consolidation

- Carefully design the materials
- Comparisons, playing with differences
- Conflict: trapping predictions
- Push reflection

Reflection turns experience into knowledge



Educational Robotics System (ERS)

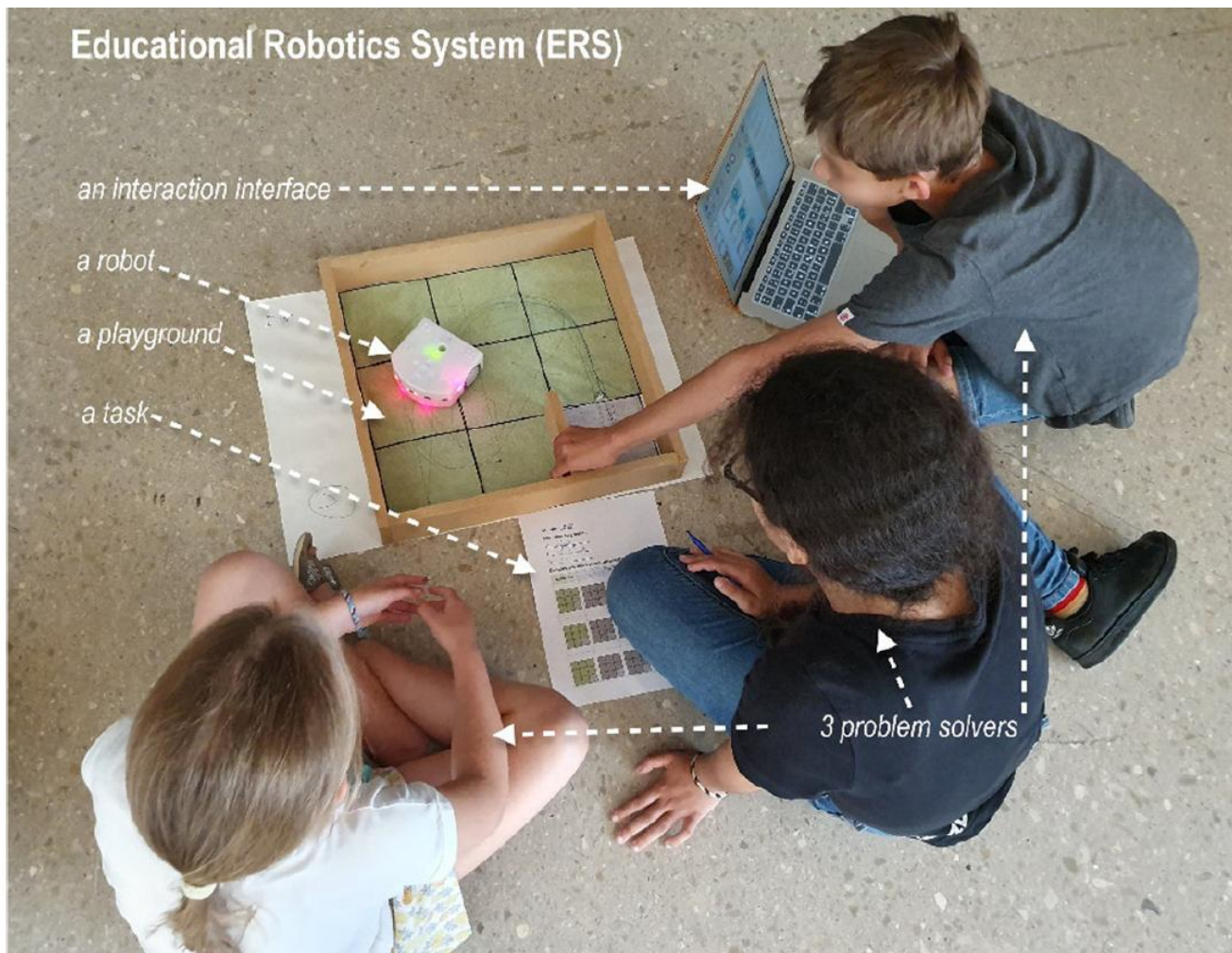
an interaction interface

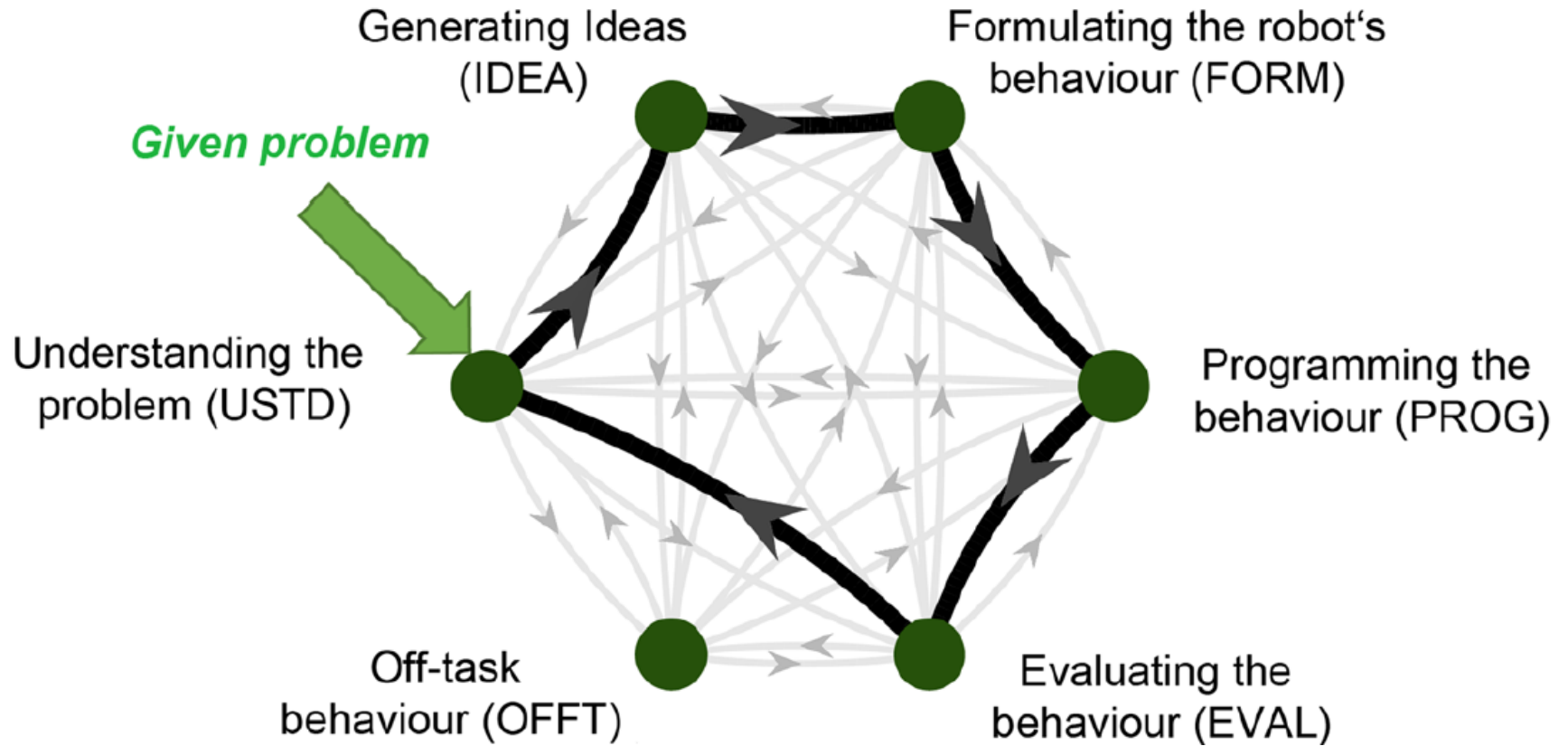
a robot

a playground

a task

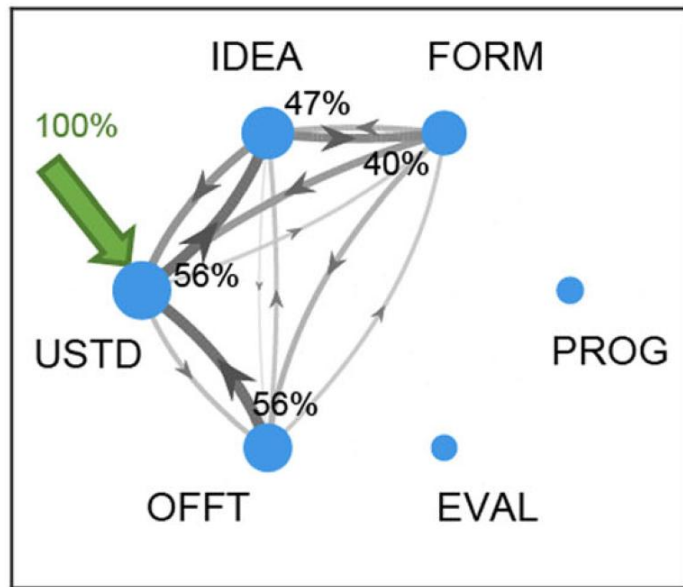
3 problem solvers





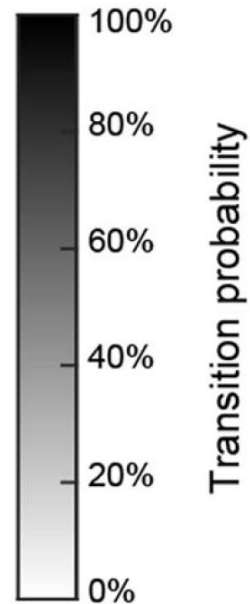
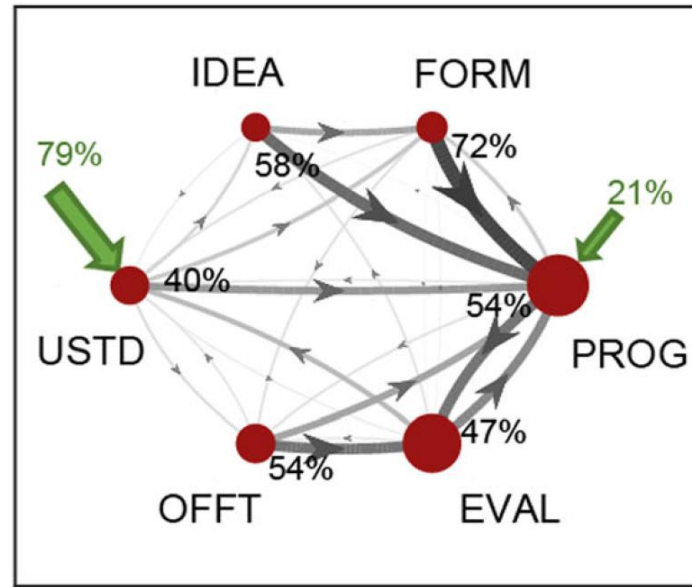
FIRST 10 MIN

Test groups

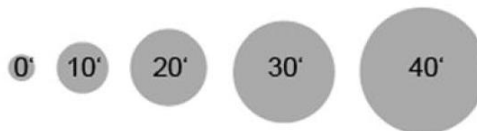


FIRST 10 MIN

Control groups



Total time per state:
(in minutes)



Remember the balloon passage?

You will see two experiments investigating people's memory for a passage. For each experiment, identify patterns you find interesting or surprising. Compare across experiments when useful. Find a way to summarize or visualize the most important patterns. For each pattern, identify the evidence in the data and **try to explain why it might occur.**

Self-Explanation

Core learning mechanism:

Learners explain new information to themselves rather than simply trying to remember it.

Why it works:

Explanations cannot state every relevant connection between ideas. Self-explanation prompts learners to fill in missing information and connections, constructing a coherent understanding that goes beyond memorization.

Example:

Explaining **why** the patterns in the Balloon Passage experiments might occur.

Guiding learners during exploration

Exploration

Consolidation

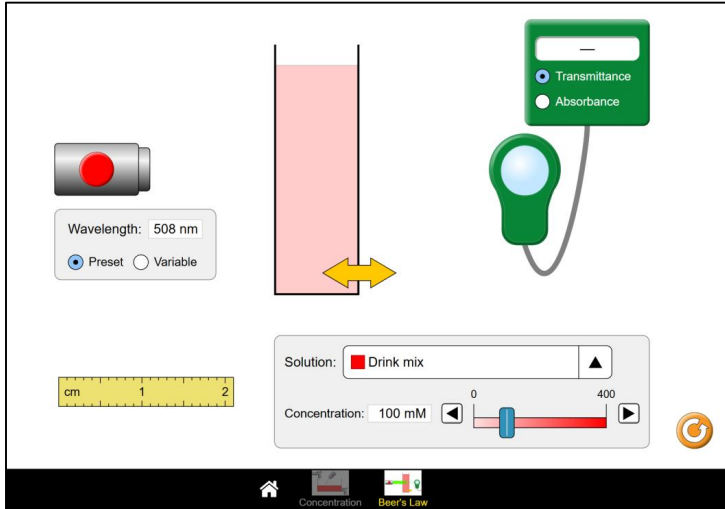
- Carefully design the materials
- Comparisons, playing with differences
- Conflict: trapping predictions
- Push reflection

Reflection turns experience into knowledge



How can we use technology to
guide learners during
exploration?

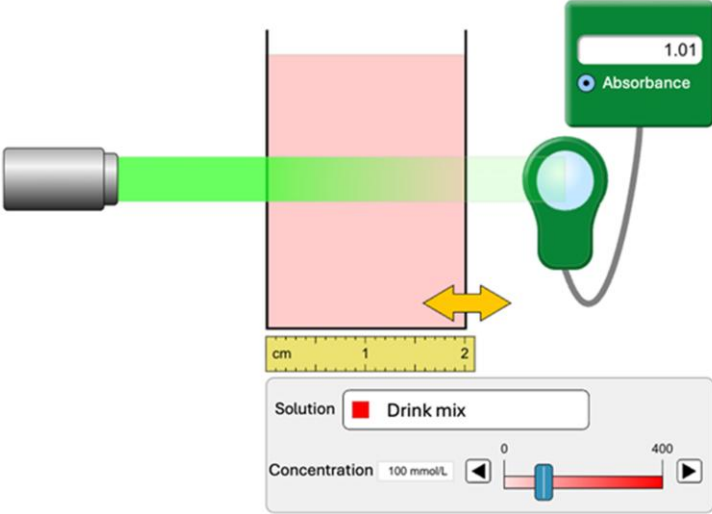
Exploration is not necessarily systematic



- Changing several variables at once
- Changing values without making systematic comparisons
- Exploring one variable at a time, but not how variables combine

Using AI to support the exploration

Simulation



Solution ■ Drink mix

Concentration 100 mmol/L

PhET

COLLECT DATA

Great! Then we'll get started right away.

What happens to the absorbance when the width doubles? Please write down your hypothesis.

I don't know. Maybe it will double in size?

Thank you — I've noted your suggestion.

Try this out in the simulation: First, set a width, read the absorbance, then double the width and read the new absorbance. Afterwards, please send me both values (width, concentration, absorbance) for both states.

(1, 340, 1.72)

Great, that's a good value! Now I need a second.

Ask Matteo...

Send

How does it work?

Prompt Structure

Role and Context:

{senior lab apprentice}

{Beer's Law Lab simulation}

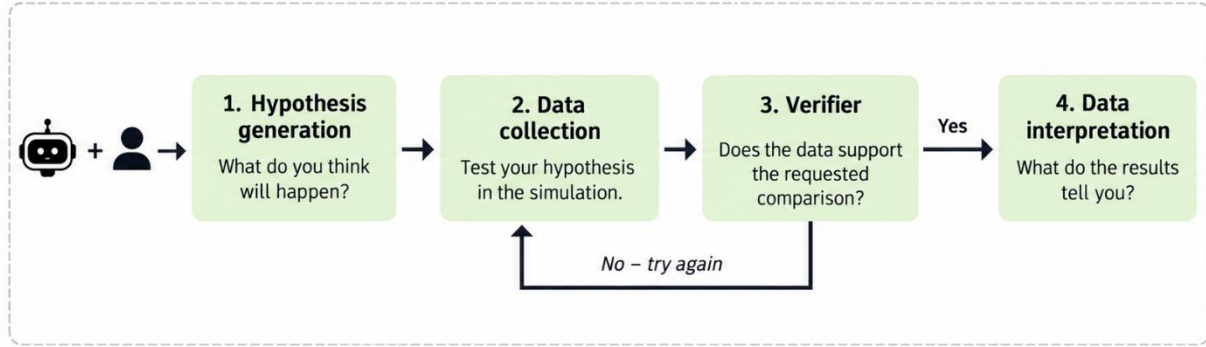
Rules:

- use a supportive, peer-like tone
- do not provide the solution or formula
- do not skip steps

Formatting:

- keep messages concise (3 short sentences max)
- one message – one question

Conversation Flow



One variable at a time

1. Width $\times 2$

2. Width $\div 3$

3. Concentration $\div 2$

4. Concentration $\times 3$

Combining variables

5. Width $\times 2$ + concentration $\times 2$

6. Width $\times 3$ + concentration $\times 3$

7. Width $\times 3$ + concentration $\div 3$

Guiding learners during exploration

Exploration

Consolidation

- **Carefully design the materials:** *Sequence of experiments*
- **Comparisons, playing with differences:** *systematic variation of width and concentration*
- **Conflict: trapping predictions:** *hypothesis before experimentation*
- **Push reflection:** *interpretation prompts after observing the data*

Take-Aways for instruction

- ① Knowledge construction**
Learning requires learners to actively build and adapt their mental representations.
- ② Explorative learning $\neq$ Leaving the learner alone**
- ③ There is a time for telling**
Exploration can prepare learners to benefit from subsequent instruction.
- ④ Learning mechanisms guide exploration**
Design materials and activities to promote comparison, conflict, and reflection.